

Agentic RAG

What is **Agentic RAG** (simple & short)

Agentic RAG = RAG + AI Agent that thinks and decides steps automatically.

First: What is RAG?

RAG (Retrieval Augmented Generation)

LLM ko external data se search karwa ke answer banwana.

Normal flow:

1. User question
2. Search vector DB
3. Get documents
4. LLM answers

Then: What is Agentic RAG?

Agentic RAG = RAG with a smart AI agent that can:

- Decide what to search
- Break question into sub-questions
- Call tools (DB, web, APIs)
- Verify answers
- Repeat steps until solved

Means **LLM becomes like a reasoning assistant, not just a chatbot.**

Simple Difference

Type	Meaning
Normal RAG	Search → Answer
Agentic RAG	Think → Plan → Search → Verify → Answer

Real Life Example (Traditional Style)

Normal RAG

You ask:

“What did Vijay say in chat 6 months ago?”

System:

- Search DB
- Return message

Agentic RAG

You ask:

“Summarize Vijay’s personality from all chats.”

Agent will:

1. Search chats
2. Group messages
3. Analyze behavior
4. Generate summary

Like a human assistant working step by step.

Why Agentic RAG Needed?

Because LLM alone:

- Has no memory
- Cannot browse
- Cannot reason multi-step reliably

Agentic RAG solves:

Long-term memory

Multi-step reasoning

Tool calling

Automation

Agentic RAG is a system where an AI agent controls retrieval and reasoning steps to answer complex questions.

Analogy (Girlfriend WhatsApp Example 😊)

Normal RAG:

You search chat manually.

Agentic RAG:

Assistant reads all chats, analyzes mood, and tells you:

“Bhai she is angry because you ignored her for 2 days.”

Companies Using Agentic RAG

- OpenAI Assistants
- LangGraph
- AutoGPT
- Devin-like AI
- Enterprise AI copilots

Normal RAG was working, so why Agentic RAG?

Normal RAG is good for simple questions.

Agentic RAG is built for complex real-world tasks.

Problems with Normal RAG (Limitations)

1.No Planning

Normal RAG only does:

Question → Search → Answer

It cannot break big tasks into steps.

2.No Multi-Step Reasoning

Example:

“Find my last 5 chats, analyze mood, and suggest a reply.”

Normal RAG: Cannot do multiple steps.

3.No Tool Control

Normal RAG can only search a database.

Cannot call APIs

Cannot run code

Cannot browse the web

4.No Verification

It answers whatever it finds.

It may give wrong answers.

5.No Long Task Handling

It cannot manage long workflows or memory tasks.

Benefits of Agentic RAG

1.Thinks Like a Human Assistant

The agent can:

- Make a plan
- Break tasks into steps
- Decide what to search

2.Solves Multi-Step Tasks

Example:

“Find all my projects, rank them, and suggest the best one.”

Agent will:

1. Search projects
2. Filter them
3. Rank them
4. Give suggestions

3.Tool Calling

Agent can:

- Query databases
- Call APIs
- Run Python code
- Use web
- Use GitHub / Notion

Normal RAG cannot do this.

4.Self-Check / Verification

Agent can:

- Generate answer
- Check again

- Improve the answer

5.Automation (Biggest Advantage)

Agentic RAG works like a mini employee.

Companies want AI that can work, not just answer questions.

Feature	Normal RAG	Agentic RAG
Search documents	✓	✓
Step-by-step reasoning	✗	✓
Use tools	✗	✓
Planning	✗	✓
Self-check	✗	✓
Automation	✗	✓

Example

Normal RAG

Librarian gives you a book.

Agentic RAG

Librarian:

- Finds books
- Reads them
- Summarizes
- Makes notes
- Gives final report